Hyperpituitarism, hypopituitarism and pituitary tumors

Block 7 – Regulation and Reproduction

2025

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

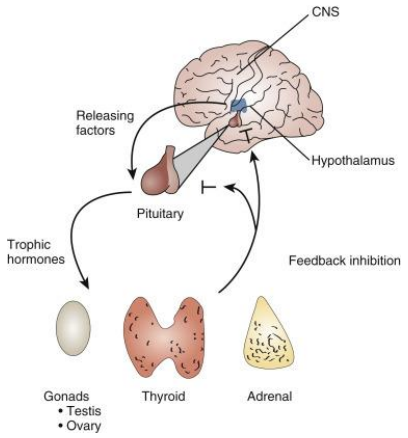
Conflict of Interest Statement

The presenters do not have a financial interest, arrangement, or affiliation with any organization that could be perceived as a real or apparent conflict of interest in the context of the subject of this presentation.

Learning Objectives

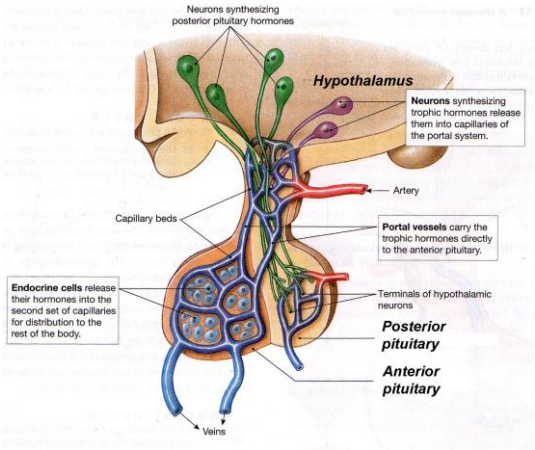
- Describe the epidemiology and signs, symptoms, and clinical features of anterior pituitary disorders.
- List clinical laboratory tests that could be utilized in identifying pituitary disorders and some of the limitations of our assays (e.g. the hook effect).
- Describe the causes and sequelae of hyper and hypopituitarism.
- Identify treatment options for anterior pituitary disorders.
- Describe neoplasms of the pituitary with emphasis on clinical, pathologic, and laboratory findings.
- Identify the causes and consequences of: a) oversecretion (e.g., syndrome of inappropriate antidiuretic hormone secretion, SIADH); and b) undersecretion (e.g., central diabetes insipidus) of vasopressin and list the principle symptoms of each
- Describe multiple endocrine neoplasia (MEN) syndromes in terms of clinical, laboratory, morphologic and genetic features.

Hypothalamic pituitary axes

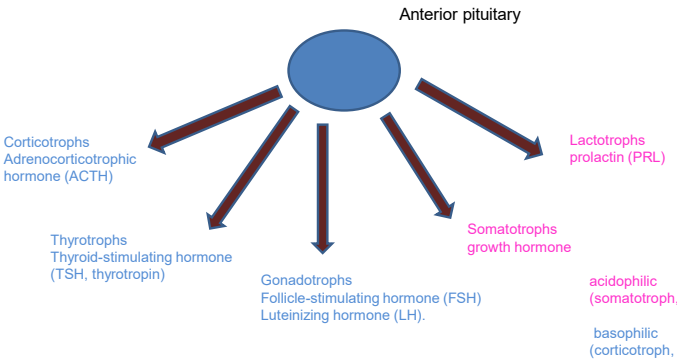


Principles of Endocrinology J. Larry Jameson, Endocrinology: Adult and Pediatric, Chapter 1, 2016

Neuro-anatomy pituitary



Anterior pituitary



acidophilic
(somatotroph, and lactotroph)

basophilic
(corticotroph, thyrotroph, and gonadotroph)

UIC UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Posterior Pituitary

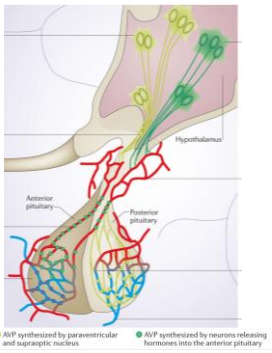
Vasopressin - Anti diuretic hormone(ADH)

Diabetes Insipidus and

SIADH (Syndrome of Inappropriate ADH)

Oxytocin

Uterine contraction, lactation, behaviors



Christ-Crain, M. et al, Copeptin in the diagnosis of vasopressin-dependent disorders of fluid homeostasis *Nat. Rev. Endocrinol. 2016*

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

7

posterior pituitary serves as a store for ADH and oxytocin made by cell bodies in the hypothalamus.
Oxytocin is mainly important for uterine contraction and lactation.
Disease states of insufficient ADH (DI) and excess ADH (SIADH) occur
Diabetes Insipidus- Deficiency of ADH leading to polyuria, hypernatremia
SIADH- Excess ADH leading to inappropriate water retention and hyponatremia

Challenges in Endocrine hormone Testing

- Pulsatility
- Assay reliability
- Lack of normative data
- Normal vs inappropriately normal

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

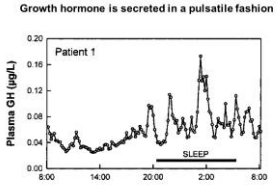
8

-Nearly all hypothalamic hormones are not measured routinely (most testing is end organ or pituitary hormone testing)
-Pulsatility- variations in GH release and different peak/trough at different times
-Reliability– degradation if not handled correctly, differences in antibody measures, low concentrations (e.g. ADH)
-Normal vs inappropriately normal– guiding principle in endo to avoid saying "normal" when in fact should be high (e.g. FSH postmenopausal)

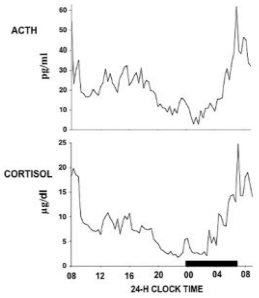
Challenges in Endocrine hormone Testing

Pulsatility

- Peak vs Trough
- Stimulate vs Suppress



From: Am J Physiol Endorinol Metab 282:943-51, 2002



Oster et al, Representative 24-hour profiles of plasma ACTH and cortisol levels in a healthy young man, The Functional and Clinical Significance of the 24-Hour Rhythm of Circulating Glucocorticoids. Endocr Rev, 2017

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

9

Two important principles in HP-end organ testing.
Peak vs trough- variable secretion at different times of day. E.g. cortisol peak early AM (useful for hypofunction testing– should be max)
Stimulate vs Suppress- general rule if concerned about hypofunction perform a stimulation test, if concerned about overproduction attempt to suppress

Challenges in Endocrine hormone Testing

Assay reliability

- Immunoassays - antibody based methods
- Mass spectrometry
- Nucleic acid based assays – PCR

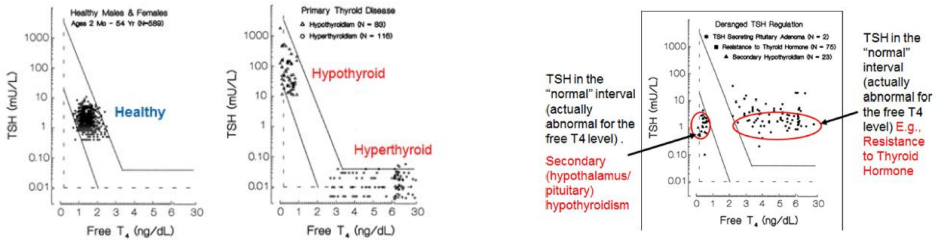
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

10

-Nearly all hypothalamic hormones are not measured routinely (most testing is end organ or pituitary hormone testing)
-Pulsatility- variations in GH release and different peak/trough at different times
-Reliability– degradation if not handled correctly, differences in antibody measures, low concentrations (e.g. ADH)
-Normal vs inappropriately normal– guiding principle in endo to avoid saying "normal" when in fact should be high (e.g. FSH postmenopausal)

Challenges in Endocrine hormone Testing

- Lack of normative data
- Normal vs inappropriately normal



Nakamoto, J, Endocrinology Laboratory Testing, Sperling Pediatric Endocrinology, 2021

Hypopituitarism – Symptoms

Depends on missing hormone

- ACTH- *fatigue, pallor, anorexia, weight loss*
- TSH- *tiredness, cold intolerance, constipation, hair loss, dry skin*
- LH/FSH-
  - Women: *oligo - amenorrhea, loss of libido, dyspareunia, infertility*
  - Men: *loss of libido, impaired sexual function, mood impairment*
- GH- *decreased muscle mass and strength, visceral obesity, fatigue, decreased quality of life, impairment of attention and memory*
- ADH- *polyuria, polydipsia, nocturia*

ACTH – adrenocorticotrophic hormone TSH – thyroid stimulating hormone LH – luteinizing hormone FSH – follicle stimulating hormone GH – growth hormone ADH – anti-diuretic hormone

Hypopituitarism - Causes

A. Neoplasms (compression)
- Intrasellar
- Parasellar
- Metastatic

B. Ischemic necrosis
- Postpartum
- Other systemic disorders

C. Vascular
- Pituitary apoplexy
- Aneurysms of ICA

D. Infectious

E. Infiltrative disease
- Hemochromatosis
- Hemosiderosis

F. Immunologic or inflammatory
- Sarcoidosis
- Granulomatous diseases

G. IatrogenicPrimary Empty Sella
- Head/Neck irradiation
- Surgical destruction

H. Metabolic disorders

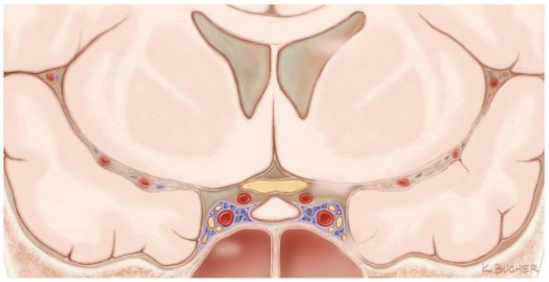
I. Genetic / Developmental Disorders

J. Traumatic

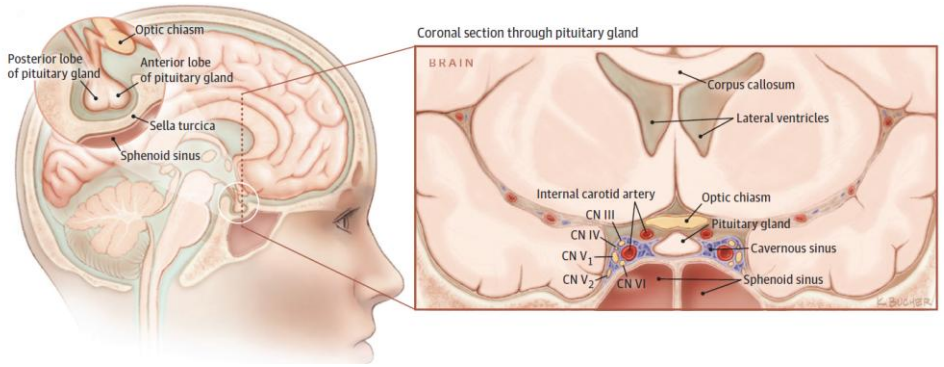
K. Idiopathic

Pituitary – Saggital and coronal sections





Pituitary – Saggital and coronal sections



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Molitch, M, Diagnosis and Treatment of Pituitary Adenomas A Review, JAMA 2017

Pituitary function testing
Assessment of pituitary – hormonal axes

Adrenal : (8 AM) Cortisol, ACTH

Thyroid : TSH, FT4

Gonadal function : (8 AM) Estradiol (females), testosterone (males), LH, FSH

Growth axis : IGF1

ACTH – adrenocorticotrophic hormone TSH – thyroid stimulating hormone LH – luteinizing hormone FSH – follicle stimulating hormone
GH – growth hormone ADH – anti-diuretic hormone, IGF1 – Insulin like growth factor 1

Pituitary function testing
Dynamic testing to assess over or under secretion

Stimulation – to assess for undersecretion

- Adrenal axis : Cosyntropin stimulation testing
- Growth hormone stimulation : Insulin, glucagon, arginine
- ADH : water deprivation testing

Suppression testing – to assess for hypersecretory conditions

- Adrenal : Dexamethasone suppression testing
- Growth hormone suppression : oral glucose tolerance test

ACTH – adrenocorticotrophic hormone TSH – thyroid stimulating hormone LH – luteinizing hormone FSH – follicle stimulating hormone GH – growth hormone ADH – anti-diuretic hormone, IGF1 – Insulin like growth factor 1

Hypopituitarism Treatment

Deficiency	Replacement
ACTH	Hydrocortisone
TSH	Thyroxine (T4)
LH/FSH	♀ Estrogen/Progesterone ♂ Testosterone hCG + rFSH GnRH (fertility)
GH	hGH
Prolactin	-
Vasopressin	Desmopressin

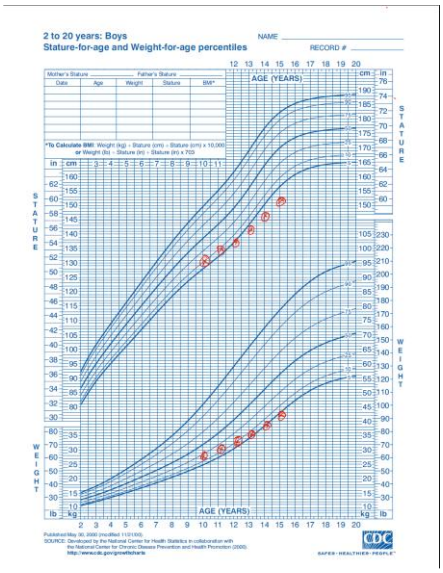
ACTH – adrenocorticotrophic hormone TSH – thyroid stimulating hormone LH – luteinizing hormone FSH – follicle stimulating hormone rFSH – recombinant Follicle stimulating hormone hCG – human chorionic gonadotropin GnRH – Gonadotropin releasing hormone GH – growth hormone ADH – anti-diuretic hormone, IGF1 – Insulin like growth factor 1

A case of delayed puberty

Case of delayed puberty

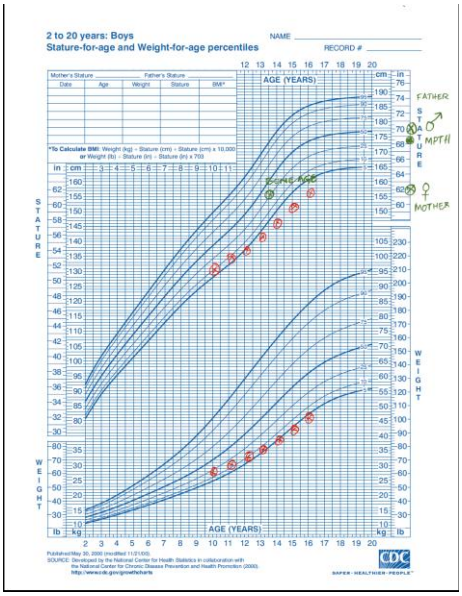
A 16 year old male presents with concerns about short stature and delayed pubertal development. Patient and his parents are worried that his peers at school are taller than him and more “masculine”.

Case of delayed puberty



UIC UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Case of delayed puberty



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Case of delayed puberty
Examination (pertinent findings)

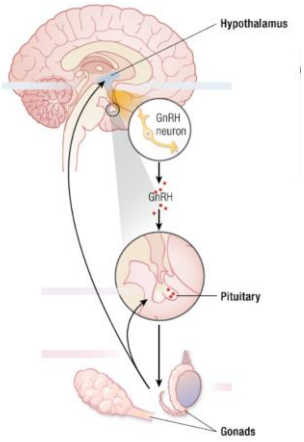
Patient well nourished, appears smaller for his age, does not have secondary sexual characters such as facial hair, deeper voice, no armpit hair,

Is noted to have some pubic hair, prepubertal genital development

Tanner stage 2 for pubic hair development

Tanner stage 1 for genital development

Hormones regulating puberty



Howard SR, et al, Delayed Puberty-Phenotypic Diversity, Molecular Genetic Mechanisms, and Recent Discoveries Endocr Rev. 2019

Case of delayed puberty Evaluation

- Follicle stimulating hormone (FSH)
  2 mIU/ml (Reference for age : 2.6-11)
- Luteinizing hormone (LH)
  0.4 mIU/ml (reference for age : 0.4-7)
- Testosterone 15 ng/dl (reference for age: 350-970)
- Prolactin 14 ng/ml (reference : 3-18)

- Bone age delayed at around 12 y at chronological age 16 y

A case of delayed growth and development

Labs noting low gonadotropins with low testosterone suggestive of

- Hypogonadotropic hypogonadism

A case of delayed growth and development

Labs noting low gonadotropins with low testosterone suggestive of

- Hypogonadotropic hypogonadism

Further questioning reveals that patient lacks a sense of smell – anosmia

Diagnosis

Kallmann syndrome

Diagnosis

Kallmann syndrome

Further evaluation ?

MRI brain, genetic studies

Diagnosis

Kallmann syndrome

MRI brain notes olfactory-bulb hypoplasia

Kallmann syndrome

- Male hypogonadism with or without anosmia - olfactogenital dysplasia
- May also have other symptoms or signs such as hearing impairment, and skeletal abnormalities, synkinesia (upper limb mirror movements), cleft lip/palate and hypodontia.

All of the following conditions are usually associated with hypogonadotropic hypogonadism except

A. Prolactinoma

B. Klinefelter syndrome

C. Kallman syndrome

D. Celiac disease

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

32

A variety of CNS insults may disrupt production of gonadotropins. The GnRH pulse generator may be disrupted by an interfering substance, such as excess prolactin (with or without hypothyroidism), or by stress, chronic illness, malnutrition, or excessive physical activity.

Klinefelter syndrome is usually associated with Hypergonadotropic Hypogonadism

POLL

All of the following conditions are usually associated with hypogonadotropic hypogonadism except

A. Prolactinoma

B. Klinefelter syndrome

C. Kallman syndrome

D. Celiac disease

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

34

A variety of CNS insults may disrupt production of gonadotropins. The GnRH pulse generator may be disrupted by an interfering substance, such as excess prolactin (with or without hypothyroidism), or by stress, chronic illness, malnutrition, or excessive physical activity.

Klinefelter syndrome is usually associated with Hypergonadotropic Hypogonadism

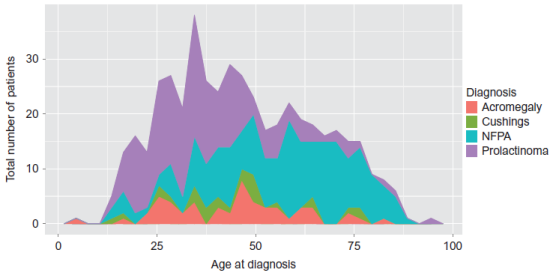
Pituitary tumors

Pituitary Tumors

- Nonfunctional
- Functional

Pituitary

- microadenomas(<10mm)
- macroadenomas (10 mm)
- giant adenomas (40 mm)

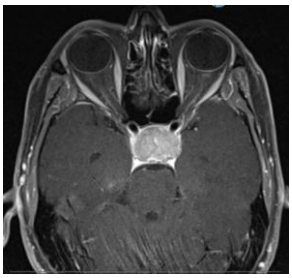


Age a diagnosis of pituitary tumor subtypes.
NFPA – Non functioning pituitary adenoma

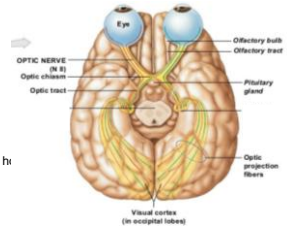
Agustsson et al., The epidemiology of pituitary adenomas in Iceland, 1955-2012: a nationwide population-based study, EJE 2015

Pituitary tumors - Mass Effects

- Hypopituitarism
  - *GH > LH/FSH > TSH = ACTH >>> ADH*
- Headache
- Vision disturbances







ACTH – adrenocorticotrophic hormone TSH – thyroid stimulating hormone LH – luteinizing hormone FSH – follicle stimulating h
GH – growth hormone ADH – anti-diuretic hormone, IGF1 – Insulin like growth factor 1

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

37

Regardless of functional status, tumors in the perisellar space are concerning for injury to important structures in this limited area.
When destruction of pituitary cells happens the deficits caused occur often in a predictable way (described Left to right in commonness).
Can think about it in terms of evolutionary adaptation– can live without GH and repro function. Others not so much.
Review optic chiasm anatomy and relation to pituitary and stalk.
Discuss compressive effect leading to hemianopsia (or first superior bitemporal quadranopsia most often)

Hypersecretion of hormones

Acromegaly - Growth Hormone Excess

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

39

Shrek actually modeled after French wrestler Maurice Tiller (The French Angel) who suffered from acromegaly

Acromegaly - Growth Hormone Excess

- Acromegaly- *Adults*
- Gigantism- *Children*

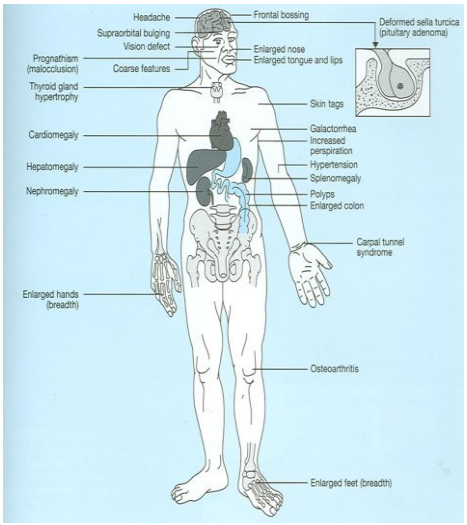


UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

40

Shrek actually modeled after French wrestler Maurice Tiller (The French Angel) who suffered from acromegaly

Acromegaly – Growth Hormone Excess



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

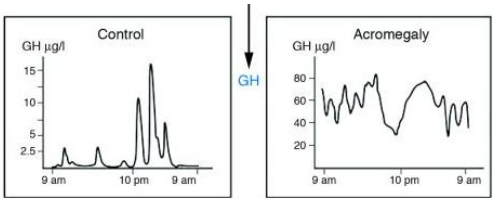
41

Growth Hormone (GH) Excess
Dysfigurement syndrome: Facial features,big/wide hands → carpal tunnel
Systemic: Sleep Apnea (Due to palate and uvula growth),
Arthropathy (enlarge cartilage, soft tissue, bones)
Organomegaly
Hypertension (very common)
Cardiomyopathy (Diastolic dysfunction -> arrythmia or CHF)
Diabetes
Neoplasia (skin tags, colon polyps[?Cancer], no evidence of increased incidence of other CA)

Acromegaly - Diagnosis

Clinical Suspicion

- ~~Random GH~~
- IGF-1



If IGF1 elevated OR equivocal

→Dynamic test – suppression testing

confirmation of the diagnosis - lack of suppression of GH to less than 1 mcg/L following documented hyperglycemia during an oral glucose load

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

42

GH is secreted in episodic nature
measure IGF-1 also
When Random GH and IGF1 levels are inconclusive – recommend OGTT GH measurements (Dynamic Test)

Acromegaly Treatment Options

- Surgery
- Medical Therapy
- Radiation Therapy

Growth Hormone-ReceptorAntagonist (GHRA)
- Pegvisomant

Somatostatin receptor ligand (SRL)
- Octreotide
- Lanreotide
- Pasireotide

Dopamine Agonist (DA)
- Cabergoline

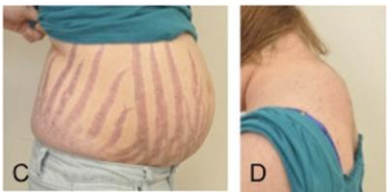
Cortisol excess - Cushing Disease / Syndrome

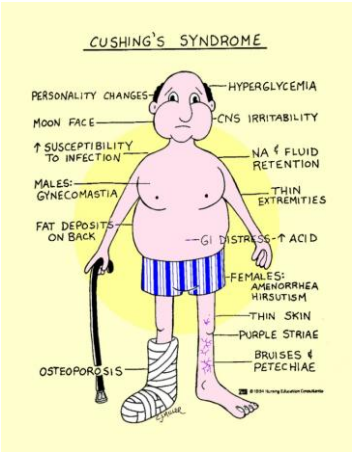
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Differentiate CD from CS. Describe clinical features a/w cortisol excess state.
Cushings Disease= ACTH secreting pituitary adenoma
Cushings Syndrome = phenotype of cortisol excess

Cortisol excess - Cushing Disease / Syndrome







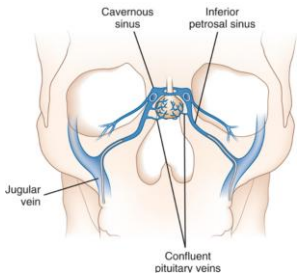
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

45

Differentiate CD from CS. Describe clinical features a/w cortisol excess state.
Cushings Disease= ACTH secreting pituitary adenoma
Cushings Syndrome = phenotype of cortisol excess

Cushing Disease- ACTH dependent hypercortisolism

- Confirm Hypercortisolism
  - *24hr urine free cortisol x 2*
  - *Salivary Cortisol x 2*
  - *Dexamethasone Suppression*
- ACTH level
- Localize
  - *MRI*
  - *High Dose Dexamethasone suppression testing*
  - *BIPSS - Inferior Petrosal Sinus Sampling and Selective Venous Catheterization*



Oldfield EH, et al. Preoperative lateralization of ACTH-secreting pituitary microadenomas by bilateral and simultaneous inferior petrosal sinus sampling. N Engl J Med. 1985

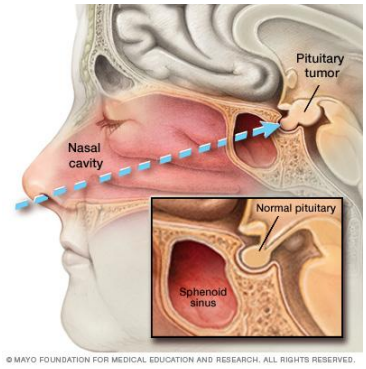
Confirm- either 24 UFC x2, salivary x2, or LDST x1 to screen, but often do at least one alternative if screen positive (e.g- UFCx2 +, then salivary)
ACTH testing to confirm ACTH-dependence (Usually either mid-high normal, or frankly high).
MRI and if 6mm or greater perform surgery. HDDST. If uncertain BIPSS to sample both sinuses compared to periphery

Cushing Disease Treatment

- Surgery
- Repeat Surgery

- Medical Therapy
- Radiation

- Adrenalectomy



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

47

Surgery primary therapy. Ideally with skilled neurosurgeon w/ high success rate.
Consider another surgery if initial not believed ideal.
Several medical therapies available.
XRT takes time to work.
Adrenalectomy always option but lifetime replacement and Nelson's syndrome risk

Cushing Disease Medical Therapy

- *Target Pituitary*
  - Somatostatin Analogues - Pasireotide
  - Dopamine Agonists - Cabergoline
- *Block GC Receptor*
  - *Glucocorticoid Receptor Antagonists* - Mifepristone (RU-486)
- *Target Adrenal glands*
- *inhibition of adrenal steroidogenesis*
  - Ketoconazole
  - Metyrapone
  - Etomidate
  - Mitotane

Syndrome of Inappropriate ADH secretion (SIADH)

Syndrome of Inappropriate ADH secretion (SIADH)

- Hyponatremia - Decreased plasma osmolality of the extracellular fluid (Posm less than 280 mOsm/kg H2O)
- Inappropriate urinary concentration during hypo-osmolality (Uosm greater than 100 mOsm/kg H2O, normal renal function)
- Clinical euvolemia
- Elevated urinary sodium excretion (>30 mmol/L) on normal salt and water intake
- Absence of other potential causes of euvolemic hypo-osmolality, particularly hypothyroidism, hypocortisolism, and diuretic use

Hannon MJ et al, Vasopressin, Diabetes Insipidus, and the Syndrome of Inappropriate Antidiuresis, Endocrinology: Adult and Pediatric, 7th ed, 2016

Syndrome of Inappropriate ADH secretion (SIADH) - *Etiologies*

- ***Tumors*** -tumors of lungs, gastrointestinal tract, Pancreas, genitourinary tract, Nasopharyngeal, Sarcoma
- ***Drugs*** - DDAVP, neuroleptics, Carbamazepine, Cyclophosphamide, Vincristine
- ***Pulmonary*** - Pneumonia (esp. mycoplasma), Tuberculosis, Positive pressure ventilation
- ***Intracranial Pathology*** – Tumors, Meningitis, Encephalitis, Absces, Vasculitis, Subarachnoid hemorrhage, Subdural hemorrhage, Traumatic brain injury
- ***Miscellaneous -*** Acute intermittent porphyria, HIV, Idiopathic

Hannon MJ et al, Vasopressin, Diabetes Insipidus, and the Syndrome of Inappropriate Antidiuresis, Endocrinology: Adult and Pediatric, 7th ed, 2016

Summary – Pituitary tumors

- Assess symptoms
- Determine if symptoms are suggestive of any particular hormone
- Are symptoms suggestive of hormone excess or deficiency
  - appropriate testing – labs, imaging
  - Treat

	Surgical	Radiation	Medical
Prolactin	Secondary	Adjunct	**Primary**
GH	**Primary**	Adjunct	Secondary
ACTH	**Primary**	Adjunct	Secondary
TSH	**Primary**	Adjunct	Secondary
FSH/LH/NF	**Primary**	Adjunct	-

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

52

Case

Chief concern

- milky nipple discharge
- headache

Case *HPI*

History of present illness (HPI)

The patient is a 26-year-old female, who states that she has been having bilateral white milky nipple discharge for several months that doesn't appear to be related to any event. She notes that she has to change her bra several times a day, due to wetness. Additionally she has been having a headache for several months, which is 5/10 in intensity without exacerbating or alleviating factors

Case *PMH/Meds/Allergies*

Past medical history : Prior parathyroidectomy for parathyroid adenoma three years previously

Gynecologic history : Menarche at age 12 years. Periods were regular every month, until the last year, since when they have been less frequent.

Married, not using any methods for contraception.

No children, but desires children, and she has been trying to conceive for one year, but her interest in sex is reduced

Medications : None

Allergies : No known allergies

Case ***SH/FMH***

Social History : Does not smoke, drink or use drugs

Family History : Father with hypertension; Mother with history of diabetes mellitus; Sister without a significant past medical history, No children

Case - ***Review of systems***

Constitutional : generalized weakness/fatigue, negative weight loss, negative for fever, chills, night sweats

HEENT : negative for congestion, rhinorrhea, blurry vision

Respiratory : negative for shortness of breath (SOB), orthopnea, cough, cyanosis

Cardiovascular : negative for tachycardia, palpitations, chest pain, pedal edema

Gastrointestinal : negative for abdominal pain, nausea, vomiting, constipation, diarrhea

Genitourinary : negative for polyuria, dysuria, hematuria, nocturia

Musculoskeletal : negative for myalgias, arthralgias, fractures, deformities

Neurological : positive headaches, negative weakness, negative seizures, negative paresthesias, negative syncope

Endocrine : irregular menstrual cycles, b/l milky breast discharge

Integumentary : negative for rash, pigmentation, lesions

Psychiatric : frustrated about symptoms

Case *Vitals*

- Blood pressure 119/76 mmHg, in both upper extremities
- Pulse is 68 beats/min
- Respiratory rate is 15 breaths/minute
- Temperature of 98°F
- BMI: 23.14 kg/m2
- Weight 60 kg (132 lbs), Height 161 cm (5 ft 3.4 in)

Case ***Physical Exam***

General : well nourished, no distress

HEENT : nasal mucosa, oropharynx and conjunctiva appear unremarkable, pupils are equally reactive to light and accommodation (PERRLA), ***Visual field testing identifies reduced vision in the bilateral outer/temporal visual fields.*** No goiter, 5 cm horizontal scar in the midline of her neck

Respiratory : lungs clear to auscultation bilaterally

Cardiovascular : regular rate and rhythm, heart sounds S1, S2 auscultated. No murmurs or added sounds noted. Peripheral pulses are felt bilaterally with no apparent jugular venous distention or peripheral edema noted

Breast : bilateral expressible milky secretions, no tenderness, no palpable masses

Gastrointestinal : non-distended in appearance, soft and non tender to palpation, bowel sounds present x 4, no bruits. No masses or organomegaly appreciated.

Neurologic : alert and oriented to time, place, person, Cranial nerves III-XII grossly intact, no ataxia, strength 5/5 in all extremities

Integumentary : no rashes, abnormal pigmentation, lesions

Musculoskeletal : No tenderness, deformities

Psychiatric : Appropriate mood and affect

Causes of Galactorrhea

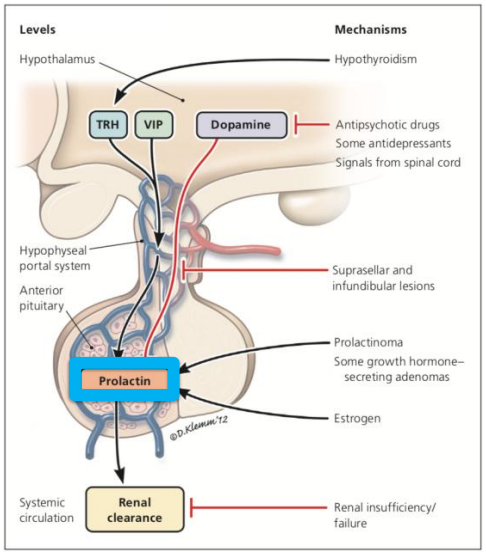
Hyperprolactinemic Galactorrhea		Normoprolactinemic Galactorrhea
Physiologic	Pathologic	
Nipple stimulation	Hypothalamic lesions	
Pregnancy	Pituitary lesions	
	Medications	
	Systemic diseases (hypothryoidism and renal failure)	

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

60

The causes of galactorrhea can be broadly categorized into two main categories, Hyperprolactinemic galactorrhea and normoprolactinemic galactorrhea. Hyperprolactinemic galactorrhea remains the most clinically significant subcategory and can be broadly categorized as physiologic including nipple stimulation and pregnancy and pathologic including hypothalamic and pituitary lesions, medications and systemic diseases. Normoprolactinemic patients with idiopathic, nonbothersome galactorrhea can be reassured and do not need treatment.

Regulation of prolactin level under physiologic and pathologic conditions



TRH = thyrotropin-releasing hormone;
VIP = vasoactive intestinal polypeptide.

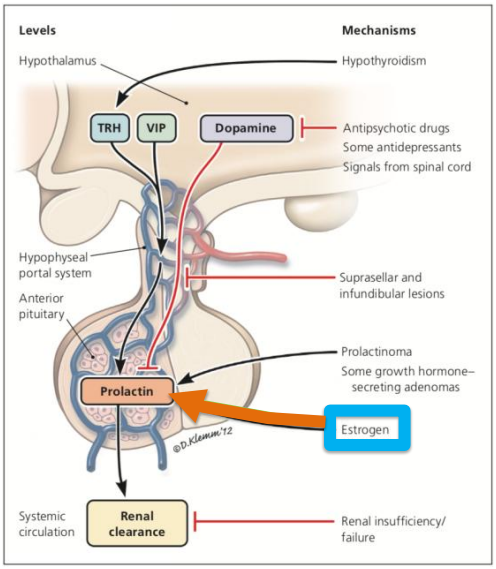
UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

61

Huang, W, et al. *Am Fam Physician*, 2012

Causes of hyperprolactenemic galactorrhea can be broadly characterized as physiologic or pathologic, as just discussed. Here is an illustration of the regulation of prolactin level under physiologic and pathologic conditions. Arrows indicate stimulation, whereas bars depict inhibition. Prolactin is essential for morphologic breast development and lactation, which includes synthesis of milk and maintenance of milk production. The regulation of serum prolactin occurs at different levels, which is illustrated on the left. The right side lists the common causes of hyperprolactinemia and relevant mechanisms. (TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.) Breast development is affected by a number of factors, including estrogen, progesterone, and prolactin (PRL) levels. However before discussing galactorrhea further, it is important to revisit the normal mechanism underlying breast development and lactation.

During pregnancy increased estrogen levels stimulate lactotroph proliferation and PRL secretion directly



Huang, W, et al. *Am Fam Physician*, 2012

TRH = thyrotropin-releasing hormone;
VIP = vasoactive intestinal polypeptide.

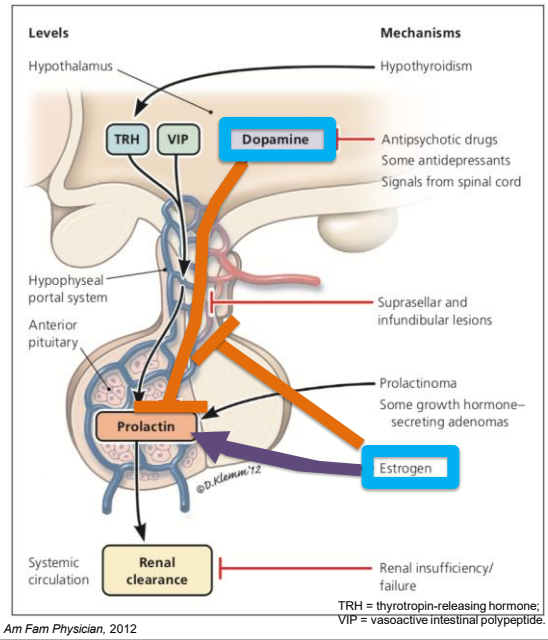
During pregnancy increased estrogen levels stimulates lactotroph proliferation and PRL secretion directly

During pregnancy increased estrogen levels stimulates lactotroph proliferation and PRL secretion directly

Estrogen also stimulates prolactin secretion indirectly by removing the inhibitory effect of the dopaminergic neurons.

Prolactin stimulates the growth and development of the mammary tissue, in preparation for the production of milk.

.



TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.

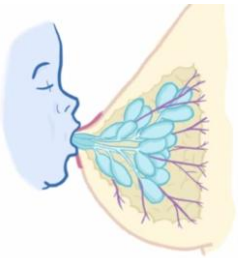
UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

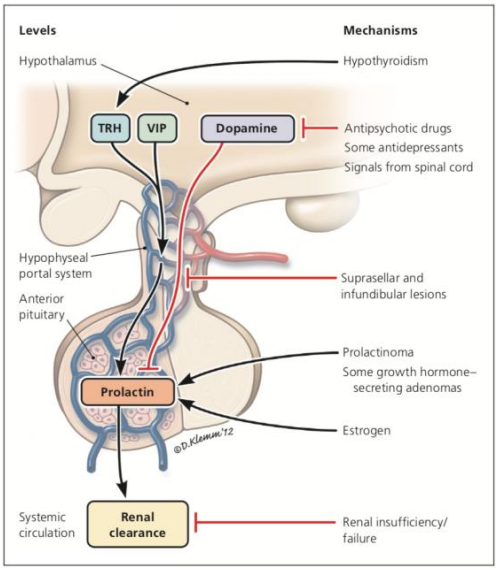
Huang, W, et al. *Am Fam Physician*, 2012

During pregnancy increased estrogen levels stimulate lactotroph proliferation and PRL secretion directly and
indirectly by inhibiting the hypothalamic dopamine neurons, thereby removing the inhibitory effect of the dopaminergic neurons. The level of prolactin in the blood increases markedly during pregnancy, and stimulates the growth and development of the mammary tissue, in preparation for the production of milk. However, milk is not normally secreted then, because progesterone and estrogen, the hormones of pregnancy, block this action of prolactin. After delivery, levels of progesterone and estrogen fall rapidly, prolactin is no longer blocked, and milk secretion begins. However during pregnancy, this process may get altered leading to premature milk production.

During pregnancy the high levels of estrogen and progesterone block milk secretion. After delivery levels of these hormones fall rapidly allowing milk secretion to begin, but without breast feeding prolactin levels will drop.

Following delivery prolactin levels increase with each suckling episode stimulating milk production.





UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

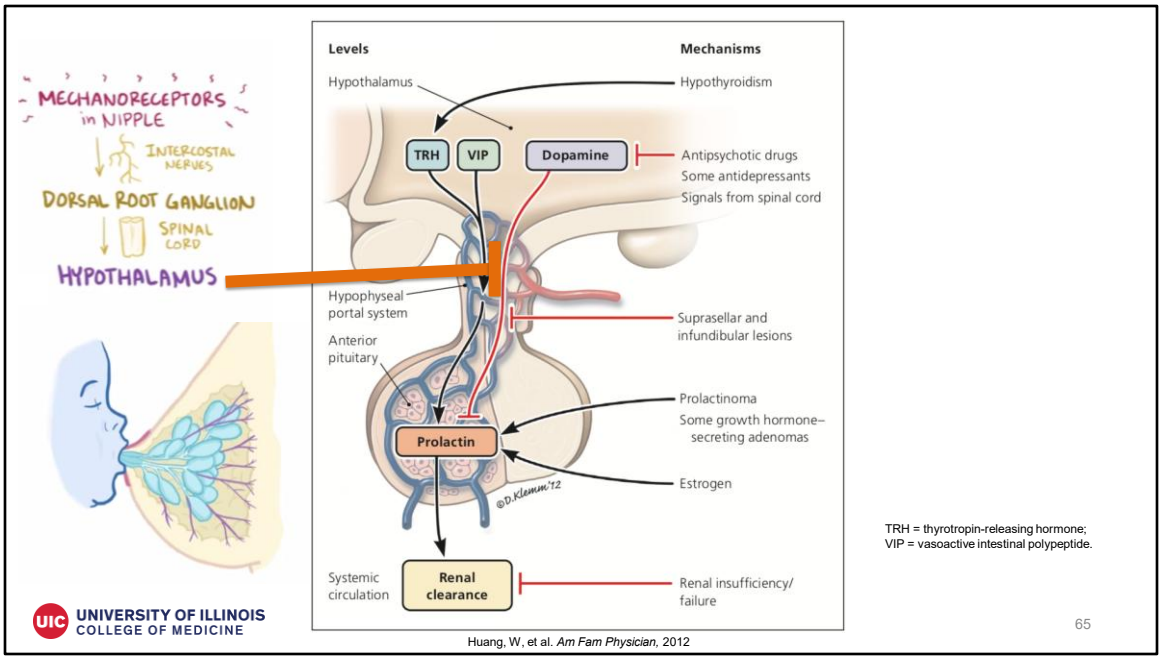
Huang, W, et al. *Am Fam Physician*, 2012

TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.

64

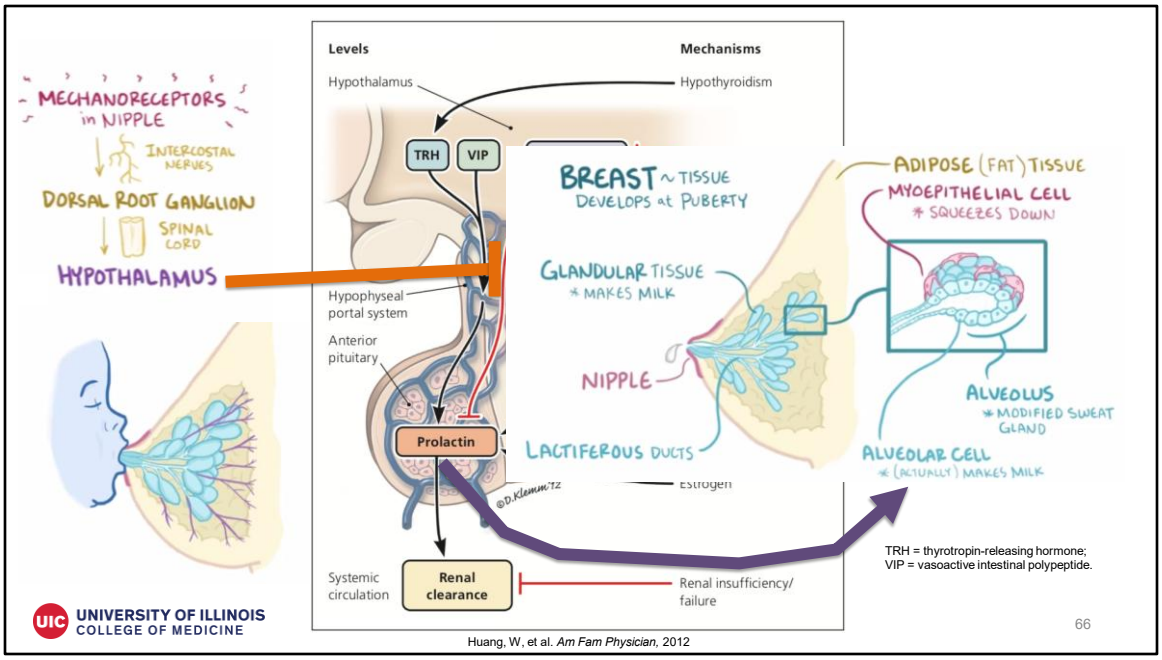
After pregnancy the drop in estrogen and progesterone levels removes the inhibitory effect at the breast, allowing milk production to begin, but without breast feeding prolactin levels will drop.

Following delivery, PRL increases with each suckling episode, stimulating milk production. Over the next several weeks, as breastfeeding gradually shifts from being on demand to a schedule, the increases in PRL levels become less. By two to three months, PRL levels no longer increase with each suckling episode, but milk production continues. Following cessation of breastfeeding, milk production rapidly decreases so that it is usually gone after a month.



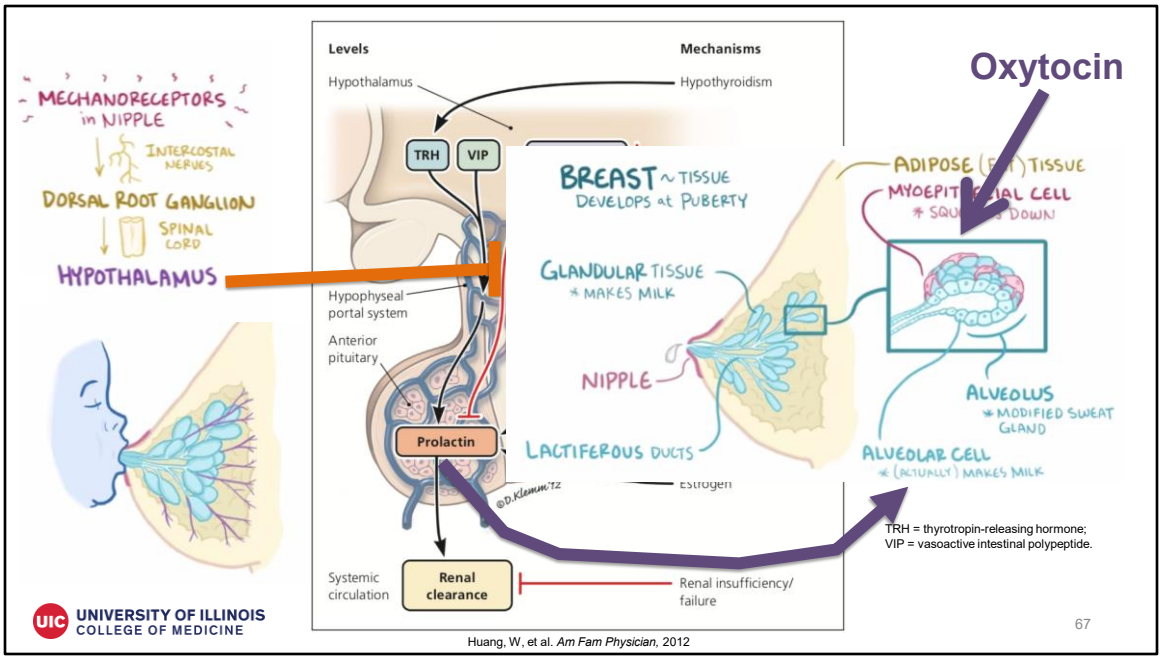
Huang, W, et al. *Am Fam Physician*, 2012

The signals from the nipples are carried via intercostal nerve fibers through the dorsal root ganglion in the spinal cord to the hypothalamus, where changes in the electrical activity of neurons that regulate the pituitary gland increase prolactin secretion, by inhibiting the dopaminergic inhibitory neurons. The suckling stimulus also triggers the release of oxytocin from the posterior pituitary gland, which triggers milk let-down. Oxytocin makes the myoepithelial cells around the alveoli contract. This makes the milk, which has collected in the alveoli, flow along and fill the ducts. Therefore prolactin controls milk production (lactogenesis) but not the milk-ejection reflex; the rise in prolactin fills the breast with milk in preparation for the next feed. Over the next several weeks, as breastfeeding gradually shifts from being on demand to a schedule, the increases in PRL levels become less. By two to three months, PRL levels no longer increase with each suckling episode, but milk production continues. Following cessation of breastfeeding, milk production rapidly decreases so that it is usually gone after a month.



Huang, W, et al. *Am Fam Physician*, 2012

These signals are carried via intercostal nerve fibers through the dorsal root ganglion in the spinal cord to the hypothalamus, where changes in the electrical activity of neurons that regulate the pituitary gland increase prolactin secretion, by inhibiting the dopaminergic inhibitory neurons. Prolactin then stimulates milk production in the breast alevolar (acinar) cells to make milk.



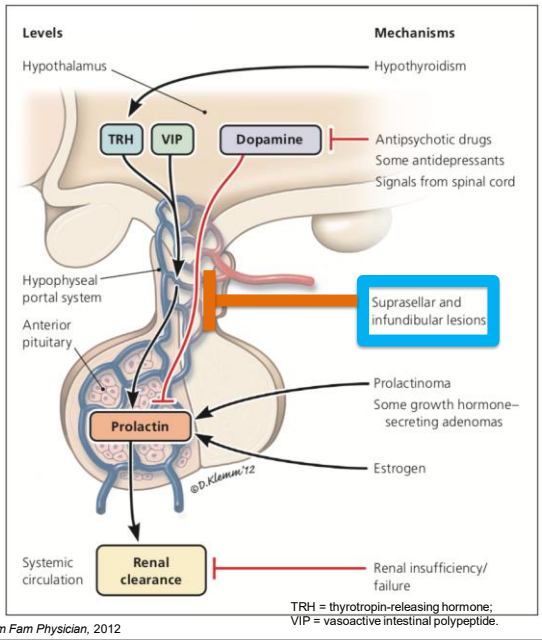
Huang, W, et al. *Am Fam Physician*, 2012

The suckling stimulus also triggers the release of oxytocin from the posterior pituitary gland, which triggers milk let-down. Oxytocin stimulates the myoepithelial cells around the alveoli contract. This makes the milk, which has collected in the alveoli, flow along and fill the ducts. Therefore prolactin controls milk production (lactogenesis) but not the milk-ejection reflex; the rise in prolactin fills the breast with milk in preparation for the next feed. Over the next several weeks, as breastfeeding gradually shifts from being on demand to a schedule, the increases in PRL levels become less. By two to three months, PRL levels no longer increase with each suckling episode, but milk production continues. Following cessation of breastfeeding, milk production rapidly decreases so that it is usually gone after a month.

Suprasellar or sellar lesions extending dorsally to involve the pituitary stalk can lead to hyperprolactinemia and galactorrhea.

This is called the "stalk effect."

Such lesions cause impairment of dopamine secretion from the hypothalamus or its transport from the hypothalamus to the pituitary gland.



TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.

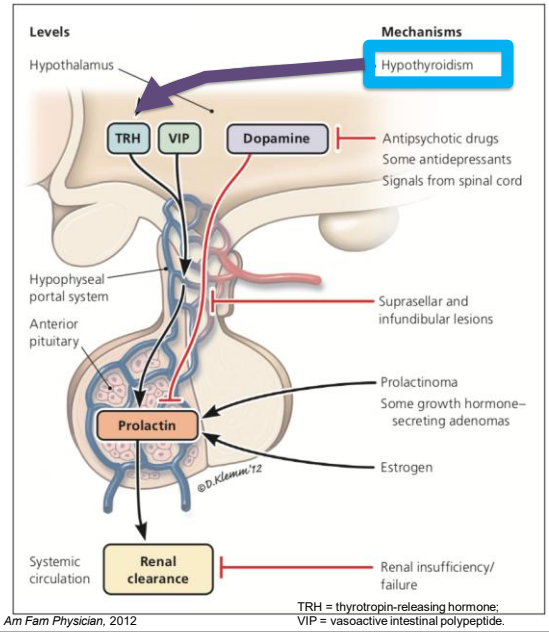
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Huang, W, et al. *Am Fam Physician*, 2012

Suprasellar or sellar lesions extending dorsally to involve the pituitary stalk can lead to hyperprolactinemia and galactorrhea. This is called the "stalk effect." Such lesions cause impairment of dopamine secretion from the hypothalamus or its transport from the hypothalamus to the pituitary gland.

Hypothyroidism can be associated with hyperprolactinemia and galactorrhea, likely because of elevated thyrotropin-releasing hormone levels resulting from less negative feedback from the thyroid hormone.

This in turn stimulates PRL secretion directly by activating thyrotropin-releasing hormone receptors in the lactotrophs or indirectly by regulating hypothalamic dopamine release.



Huang, W, et al. *Am Fam Physician*, 2012

TRH = thyrotropin-releasing hormone;
VIP = vasoactive intestinal polypeptide.

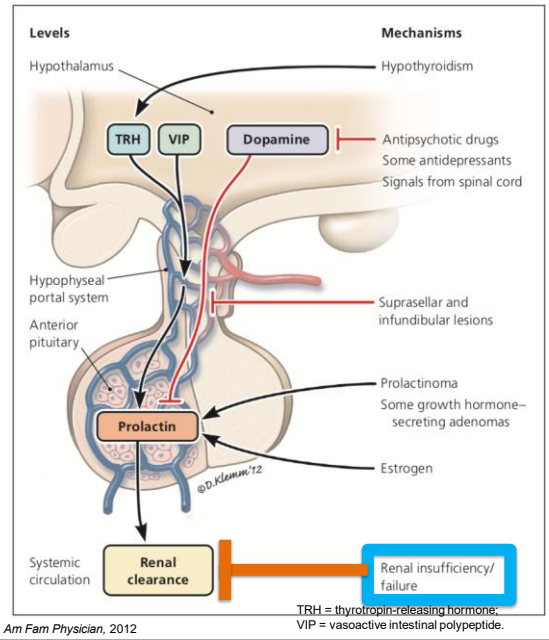
UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Hypothyroidism has been associated with hyperprolactinemia and galactorrhea, likely because of elevated thyrotropin-releasing hormone levels resulting from less negative feedback from the thyroid hormone. This in turn stimulates PRL secretion directly by activating thyrotropin-releasing hormone receptors in the lactotrophs or indirectly by regulating hypothalamic dopamine release.

Hyperprolactinemia can also occur in patients with renal insufficiency.

Prolactin levels up to 1,000 mcg/L have been reported. The mechanism is likely a result of decreased clearance of prolactin, as well as continued secretion.

After renal transplantation, hyperprolactinemia and galactorrhea can usually be corrected or significantly improved, sometimes within days.



TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.

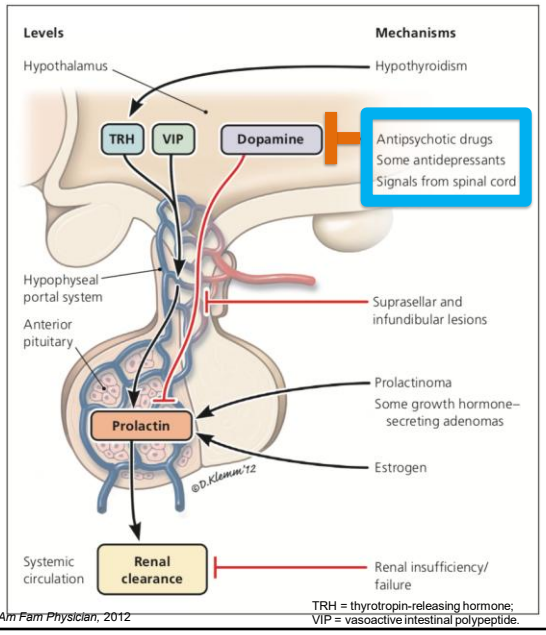
Huang, W, et al. *Am Fam Physician*, 2012

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Hyperprolactinemia can also occur in patients with renal insufficiency. Prolactin levels up to 1,000 mcg per L (43,478.00 pmol per L) have been reported. The mechanism is likely a result of decreased clearance of prolactin, as well as continued secretion. After renal transplantation, hyperprolactinemia and galactorrhea can usually be corrected or significantly improved, sometimes within days.

Hyperprolactinemia and galactorrhea are often caused by medications, most commonly antipsychotics, gastrointestinal promotility agents, and verapamil.

Antipsychotics and gastrointestinal promotility agents block the dopamine receptor on the pituitary lactotrophs, decreasing the normal inhibition of PRL release and leading to hyperprolactinemia.



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Huang, W, et al. *Am Fam Physician*, 2012

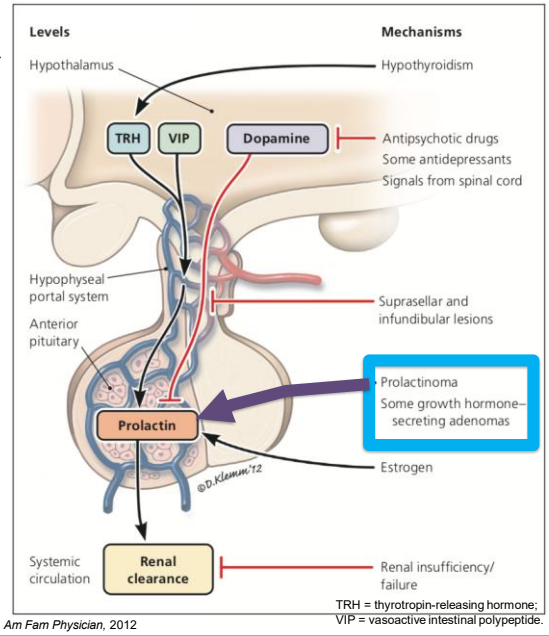
TRH = thyrotropin-releasing hormone;
VIP = vasoactive intestinal polypeptide.

Hyperprolactinemia and galactorrhea are often caused by medications, most commonly antipsychotics, gastrointestinal pro- motility agents, and verapamil. Antipsychotics and gastrointestinal promotility agents block the dopamine receptor on the pituitary lactotrophs, decreasing the normal inhibition of PRL release and leading to hyperprolactinemia.

Pituitary adenomas are the most common pathologic cause of hyperprolactinemia, which results from either direct production by the adenoma (prolactinoma) or the stalk effect.

Prolactinomas are the most common pituitary adenomas and are classified into two types based on size: microadenomas (less than 1 cm) and macroadenomas (1 cm or greater).

Circulating PRL levels usually parallel tumor size, such that microadenomas rarely result in PRL levels higher than 200 mcg/L while macroadenomas usually result in PRL levels higher than 200 mcg/L.



TRH = thyrotropin-releasing hormone; VIP = vasoactive intestinal polypeptide.

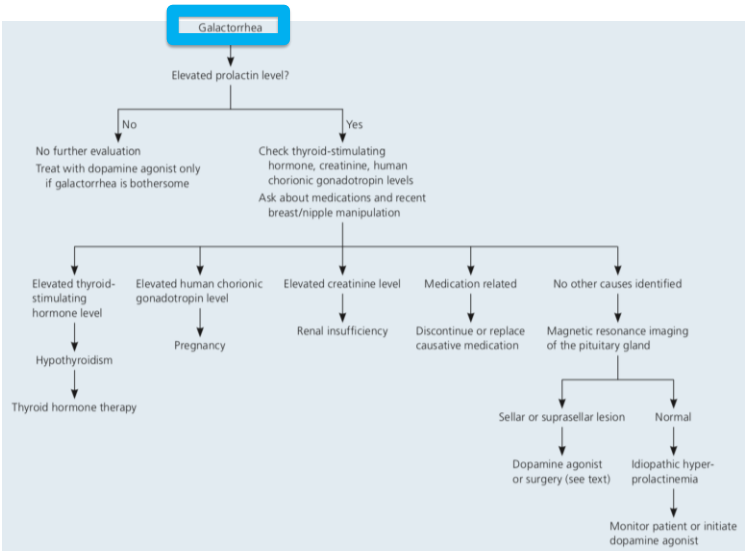
UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Huang, W, et al. *Am Fam Physician*, 2012

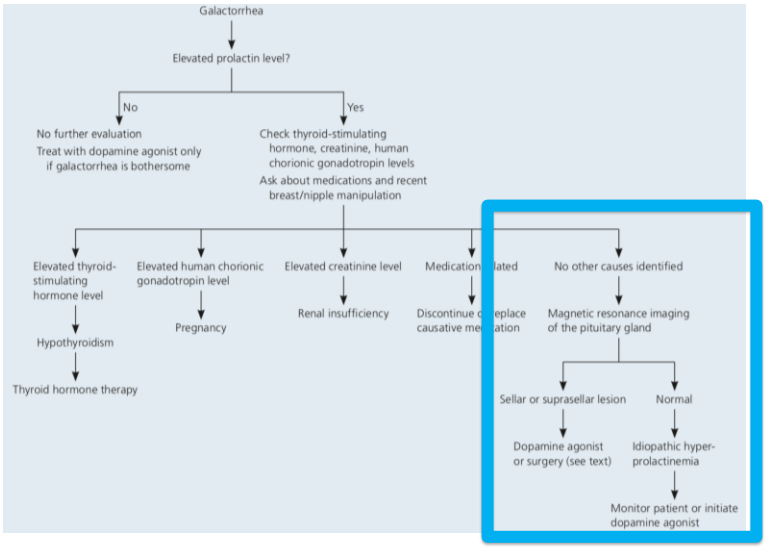
Pituitary adenomas are the most common cause of hyperprolactinemia, which results from either direct production by the adenoma (prolactinoma) or the stalk effect. Prolactinomas are the most common pituitary adenomas and are classified into two types based on size: microadenomas (less than 1 cm) and macroadenomas (1 cm or greater). Circulating PRL levels usually parallel tumor size, such that microadenomas rarely result in PRL levels higher than 200 mcg per L and macroadenomas usually result in PRL levels higher than 200 mcg per L.

Patient's labs

Lab	Value	Reference
Prolactin	**270 ng/ml**	1.2 – 29.9 ng/ml
Beta HCG (urine)	neg	
Thyroid stimulating hormone (TSH)	2.2 mIU/L	0.3 – 5 mIU/L
Free thyroxine (fT4)	0.9 ng/dl	0.7-1.9
Creatinine	0.78 mg/dl	0.6-1.00
AST	15 U/L	5-34 U/L
ALT	16 U/L	0-55 U/L
Luteinizing hormone (LH)	0.6 IU/L	1.6 – 12.4 IU/L
Follicle stimulating hormone (FSH)	2.2 IU/L	1.2 – 9.6 IU/L
Estradiol	13 pg/ml	15-350 pg/ml



Huang, W, et al. *Am Fam Physician*, 2012



UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

75

Huang, W, et al. *Am Fam Physician,* 2012

How would you interpret these results in a patient with galactorrhea and a 3 cm pituitary mass ?

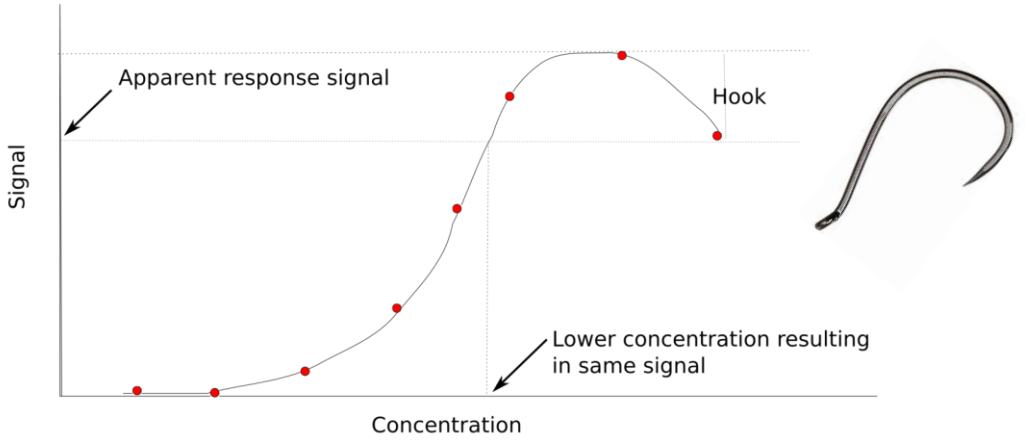
Test	Results
Prolactin	Within normal limits
Thyroid stimulating hormone	Within normal limits
Creatinine	Within normal limits
Beta-HCG	Negative

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

76

Do you trust these results? Answer: No, as prolactin should be elevated.

The “Hook Effect”



UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Schiettecatte, J et al, Interferences in Immunoassays, Advances in Immunoassay Technology, IntechOpen, March 2012.

77

The “normal” prolactin result could likely be false negative because of the hook effect. The hook effect or the prozone effect is a type of interference noted in certain immunoassays and nephelometric assays, resulting in false negatives or inaccurately low results, resulting in a characteristic hook to the graph.

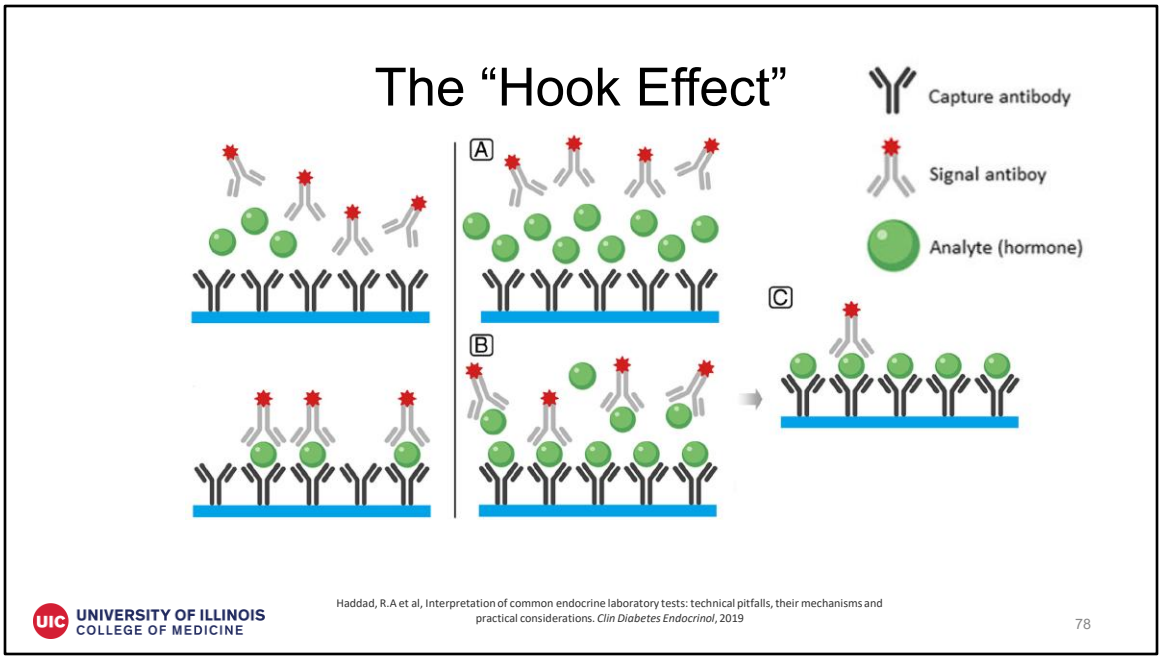
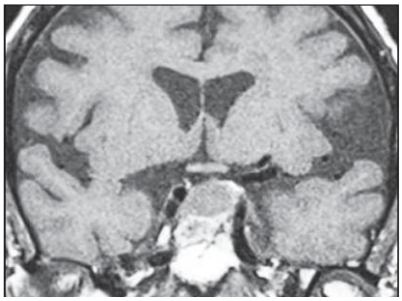
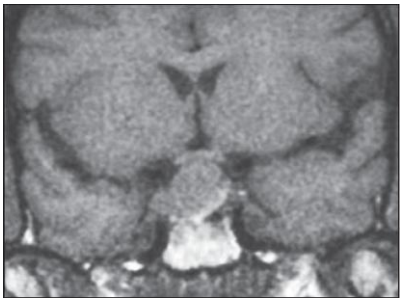


Illustration of the high dose hook effect.
The left panel illustrates the non-competitive “sandwich” immunoassay with normal (or elevated within the tolerance of the assay kit) hormone concentration.
The right panel illustrates the mechanism of the hook effect with exceedingly high hormone concentration.
a The sample that contains remarkably elevated hormone concentration is added to the test tube which contains both capture and signal antibodies. b The studied hormone overwhelmingly saturates both the capture and signal antibodies preventing the formation of the “sandwiches”. c After the washout phase, only a few “sandwiches” will be left producing a low signal

Which imaging study result, do you think is present in this patient?



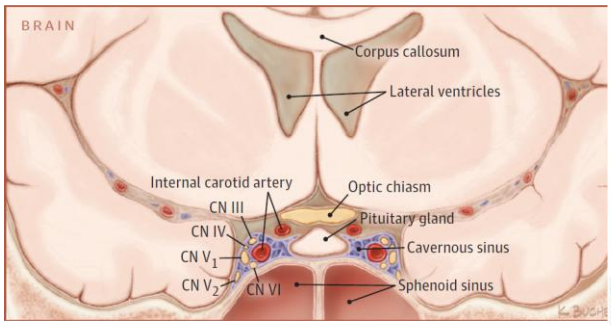


UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

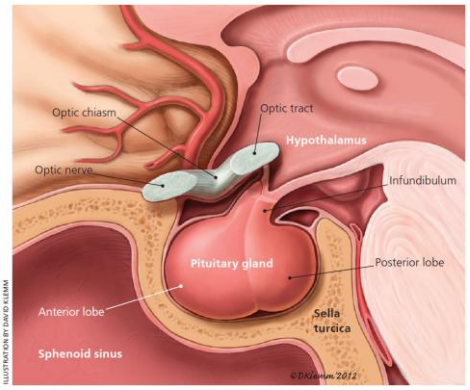
Lee, I H et al, Visual Defects in Patients With Pituitary Adenomas: The Myth of Bitemporal Hemianopsia, AJR 2015

79

How are the patients visual fields on physical examination? Answer: There is bilateral temporal field deficits; therefore look at the pituitary.

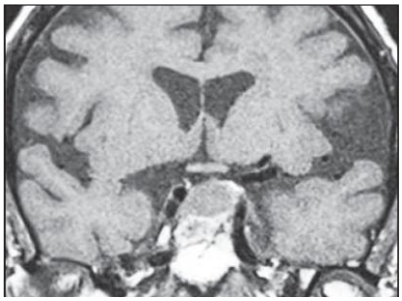


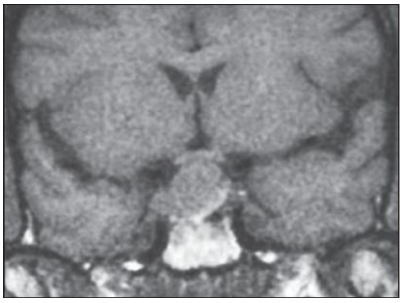
Molitch, M, Diagnosis and Treatment of Pituitary Adenomas A Review, JAMA 2017



Lake MG, Krook LS, Cruz SV. Pituitary adenomas: an overview. Am Fam Physician. 2013

Which imaging study result, do you think is present in this patient?

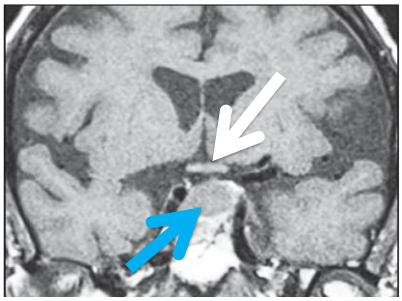


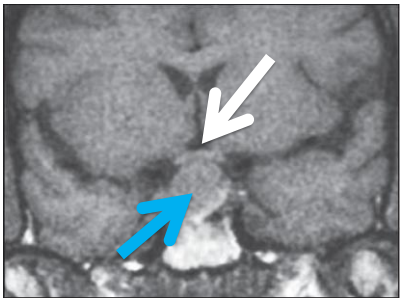


UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Lee, I H et al, Visual Defects in Patients With Pituitary Adenomas: The Myth of Bitemporal Hemianopsia, AJR 2015

Which imaging study result, do you think is present in this patient?





White arrow – optic chiasma
Blue arrow – pituitary adenoma

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Lee, I H et al, Visual Defects in Patients With Pituitary Adenomas: The Myth of Bitemporal Hemianopsia, AJR 2015

82

The correct answer is B. Visual field testing in this patient identifies reduced vision in the bilateral outer/temporal visual fields. Both images identify demonstrate pituitary tumors highlighted by the blue arrows; however only image B demonstrates tumor in contact with the optic chiasm, highlighted by the white arrows.

MR image on the left shows no contact between optic pathway and pituitary macroadenoma.
MR image on the right shows less than 3 mm of displacement (i.e., mild displacement) of optic pathway caused by compression by pituitary macroadenoma.

How are you going to treat this patient?

A. Dopamine agonist
B. Surgery
C. Watchful waiting
D. Radiation

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

How would this patient's desire for pregnancy change your management? have to consider potential teratogenic effects of medications

POLL

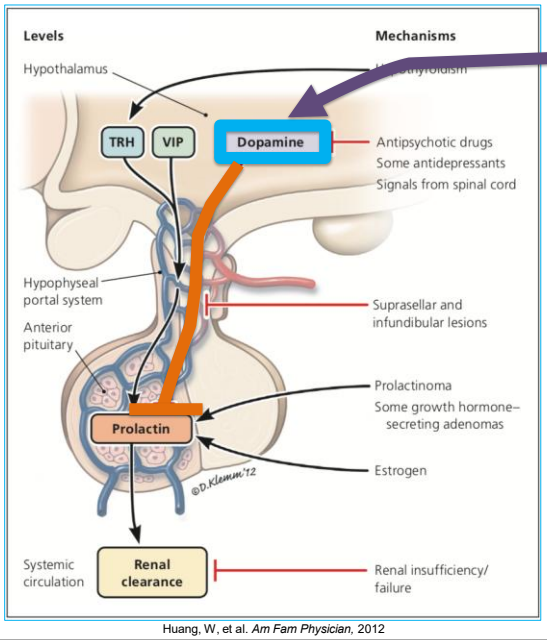
How are you going to treat this patient?

A. Dopamine agonist

B. Surgery

C. Watchful waiting

D. Radiation



Bromocriptine and Cabergoline

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

Huang, W, et al. *Am Fam Physician*, 2012

86

Bromocriptine and cabergoline, primarily function as dopaminergic agonists, which inhibit prolactin secretion; therefore, lowering prolactin levels.

Patient update

- The patient is started on cabergoline and her prolactin levels are monitored periodically with prolactin levels and MRI pituitary
- She has normalization of her prolactin levels in about six months after starting treatment and resolution of her galactorrhea
- Her headaches resolve and she has resumption of normal monthly menstrual cycles
- MRI pituitary shows significant reduction in her tumor volume and she has resolution of her bitemporal visual field deficit

Case Summary

- Hyperprolactinemic Galactorrhea
  - Pituitary adenoma (Prolactinoma)
    - Hook effect with macroadenomas
- Treatment - medical therapy
  - Dopamine agonist therapy
  - Monitor for tumor growth during pregnancy

UIC UNIVERSITY OF ILLINOIS COLLEGE OF MEDICINE

88

Today we discussed the concept of galactorrhea, which is nonlactional milk production. When galactorrhea is accompanied by amenorrhea, it is usually caused by hyperprolactinemia. Pituitary adenomas are the most common cause of hyperprolactinemia, which results from either direct production by the adenoma (prolactinoma) or the stalk effect. We also discussed the importance of recognizing potential false negatives in macroadenomas due to the hook effect. For patients with symptomatic prolactinomas, medical therapy is the treatment of choice. Dopamine agonists have been shown to be effective in normalizing prolactin, restoring reproductive hormones, shrinking adenomas, and increasing bone density via normalization of estrogen/testosterone levels, with bromocriptine being the preferred therapy choice for women trying to become pregnant. Close monitoring during pregnancy is indicated due to the potential growth of the tumor due to increased estrogen levels. Lastly we discussed multiple endocrine neoplasia syndromes, and in particularly one needs to be aware of the association of pituitary adenomas and MEN1 and MEN4.

Case - A 33 year old male with kidney stones

33 year old male with kidney stones

A middle-aged male presents to the Emergency room with concern of acute flank pain and nausea

Evaluation notes

- Ca 13.3 mg/dl (9-11)
- Urinalysis noting hematuria
- CT scan of his abdomen notes a left renal calculus

33 year old male with kidney stones

A middle-aged male with acute flank pain and nausea

Evaluation of his hypercalcemia –reveals parathyroid adenoma

33 year old male with kidney stones

A middle-aged male with acute flank pain and nausea

- Further history notes that in his mid 20's he had a resection of a pituitary tumor.
- Family history is significant for an older brother with renal stones and history of a "stomach" tumor, mother with history of "brain tumor"

What test is most likely to be diagnostic in this scenario?

A) *RET* proto-oncogene

B) testing their drinking water

C) *MEN 1* gene

D) assess for DNA repair defect

POLL

What test is most likely to be diagnostic in this scenario?

A) *RET* proto-oncogene

B) testing their drinking water

C) *MEN 1* gene

D) assess for DNA repair defect

Multiple endocrine neoplasia MEN

MEN1 - Wermer syndrome - autosomal dominant disorder - *MEN1* gene

Parathyroid, ***P***ituitary, and gastro-duodeno*P*ancreatic neuroendocrine tumors (NETs), other tumors – adrenal, bronchopulmonary

Multiple endocrine neoplasia MEN

- **MEN2A** – Sipple syndrome- autosomal dominant disorder - *RET* proto-oncogene
- medullary thyroid carcinoma(MTC), pheochromocytoma, parathyroid tumors

MEN3 or MEN2B – *RET* proto-oncogene

- MTC, pheochromocytoma, with a marfanoid habitus, mucosal neuromas, medullated corneal fibers, and intestinal autonomic ganglion dysfunction leading to megacolon

Multiple endocrine neoplasia MEN

MEN4 - autosomal dominant mutation - *CDKN1B* gene

Parathyroid adenoma, pituitary tumors, other tumors – testicular, adrenal

Questions?